

NALUT, Libyan Arab Jamahiriya

WMO: 620020

Lat: 31.867N

Lon: 10.983E

Elev: 2037

StdP: 13.65

Time Zone: 2.00 (E02)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.7	42.4	15.2	12.8	60.0	18.1	14.7	59.8	32.4	50.0	27.7	48.6	11.5	270	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	104.1	68.5	101.1	68.0	98.1	67.4	76.0	92.5	73.7	88.9	72.0	86.8	8.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	126.5	82.3	69.7	117.6	79.9	67.7	110.0	78.8	41.0	93.0	38.7	88.8	37.1	86.4	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.6	22.1	19.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.8	52.3	53.7	60.6	66.6	74.1	80.8	84.8	84.5	80.5	71.9	61.2	53.9
	DBStd	13.43	4.09	5.82	7.32	7.76	7.07	7.58	6.36	4.98	5.99	6.45	6.52	5.81
	HDD50	68	20	23	4	0	0	0	0	0	0	0	1	20
	HDD65	1493	395	320	180	73	8	1	0	0	0	14	154	348
	CDD50	6940	90	127	332	497	746	925	1078	1070	916	680	336	143
	CDD65	2887	0	4	44	119	289	475	613	605	466	228	39	5
	CDH74	33874	0	30	319	1131	2973	5985	8327	7885	5315	1729	164	16
	CDH80	17891	0	8	86	442	1346	3357	4970	4418	2680	558	26	0
Wind	WSAvg	9.2	9.7	9.5	9.9	9.9	10.2	9.4	8.8	8.3	8.7	7.8	8.6	9.8
Precipitation	PrecAvg	4.8	0.4	0.6	1.1	0.6	0.1	0.1	0.0	0.0	0.3	0.7	0.7	0.7
	PrecMax	8.9	2.8	3.2	9.9	3.2	0.6	0.6	1.1	0.2	2.0	4.4	3.4	3.7
	PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.5	0.7	1.7	0.9	0.2	0.2	0.2	0.0	0.5	1.0	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.6	78.1	86.7	93.7	98.9	106.8	108.7	104.6	102.0	91.8	83.7	76.3
		MCWB	49.3	54.0	55.1	58.8	64.9	69.7	71.1	68.3	69.5	64.5	60.8	53.7
	2%	DB	64.8	71.1	81.7	89.2	94.5	102.9	105.1	101.5	98.2	89.2	78.5	70.4
		MCWB	48.1	51.0	56.3	58.6	63.9	66.1	69.1	69.0	67.8	63.4	58.5	52.0
	5%	DB	61.7	66.9	76.8	85.3	91.3	98.9	101.5	98.6	95.4	86.2	75.0	66.3
		MCWB	47.1	50.4	55.3	57.9	63.0	66.4	68.8	67.9	67.1	62.9	56.9	50.7
	10%	DB	59.4	63.5	72.7	80.7	87.5	95.0	97.9	95.6	92.3	82.8	71.7	62.8
		MCWB	46.7	48.8	53.3	57.0	62.2	65.2	67.7	67.6	66.7	61.5	56.2	49.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.3	58.7	64.3	64.8	72.6	76.7	78.8	80.0	76.1	70.7	66.8	57.5
		MCDB	59.0	68.7	78.6	83.8	90.4	95.3	101.1	92.7	90.8	82.2	73.7	68.6
	2%	WB	51.3	54.5	60.0	61.7	68.1	72.7	74.9	76.2	73.5	68.3	63.1	54.1
		MCDB	58.2	67.3	73.2	79.6	86.7	91.8	92.3	90.9	85.5	81.2	72.7	65.1
	5%	WB	49.7	52.1	57.1	59.7	66.1	70.1	72.6	73.8	71.9	66.5	60.6	52.3
		MCDB	56.6	63.0	70.4	78.8	83.4	88.5	90.6	87.5	84.5	79.6	69.0	61.6
	10%	WB	48.6	50.1	54.9	57.8	64.4	68.0	70.5	71.8	70.4	64.7	58.6	51.0
		MCDB	55.6	60.1	68.4	76.6	81.3	85.4	88.2	85.4	83.2	76.0	67.6	59.4
Mean Daily Temperature Range	5% DB	MDBR	12.0	13.9	15.8	18.3	20.0	20.5	20.0	18.6	17.7	14.8	13.0	12.0
		MCDBR	15.3	18.6	21.1	23.9	23.1	23.3	22.7	21.1	21.1	18.7	16.7	16.1
		MCWBR	7.3	9.8	9.2	8.4	8.8	8.2	8.9	8.0	8.3	7.8	7.5	8.4
	5% WB	MCDBR	11.7	16.9	17.8	20.7	21.0	21.5	22.2	20.0	19.8	15.9	13.1	13.8
		MCWBR	6.8	10.4	9.3	8.7	9.2	9.4	10.0	10.2	8.9	8.1	7.7	8.2
Clear-Sky Solar Irradiance	taub	0.343	0.368	0.455	0.510	0.533	0.489	0.486	0.479	0.502	0.454	0.387	0.350	
	taud	2.215	2.122	1.871	1.759	1.728	1.850	1.867	1.899	1.849	1.941	2.086	2.194	
	Ebn at Noon	274	278	260	251	246	256	256	256	243	245	253	263	
	Edh at Noon	36	44	61	71	75	66	64	61	62	52	41	36	
All-Sky Solar Radiation	RadAvg	1118	1401	1767	2088	2294	2477	2501	2284	1903	1509	1180	1009	
	RadStd	59	58	80	81	101	70	52	39	64	79	48	58	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page